

SeokHyeon (Lucas) Kong

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RESEARCH INTERESTS

Quantum Computer Architecture, Quantum Compiler, Quantum Error Correction and Detection, Quantum Controller, Quantum Programming Language, Quantum Algorithm Optimization

EDUCATION

Sungkyunkwan University (SKKU)

College of Information and Communication Engineering

Bachelor of Science in Electronic and Electrical Engineering

Bachelor of Science in Advanced Semiconductor Engineering

Suwon, South Korea

Expected February 2026

GPA: 3.95/4.0 (Major: 4.0/4.0)

System Architecture Track

The Pennsylvania State University

School of Electrical Engineering and Computer Science (EECS)

Exchange Student in Computer Engineering

University Park, PA

Fall 2024

Grade: 4.0/4.0

PUBLICATIONS & PRESENTATION

S. H. Kong, D. Kim, K. Maeng*, and E. Seo*, “Characterizing the System Overhead of Discrete Gaussian Noise Generation for Differential Privacy,” *IEEE Computer Architecture Letters*, under review, 2025

S. H. Kong, C. Park, J. Moon, and D. Dahl*, “Solving Constraint Satisfaction Problems with Variational Quantum Algorithm on NISQ Devices,” *Poster presented at the Quantum Information Research Support Center (Qcenter)*, Suwon, South Korea, Sep. 2025

S. H. Kong and H. Oh*, “Predicting Key Regional Real Estate Prices Using Machine Learning Technique: with an emphasis on Jeonsae system,” *Journal of the Korean Institute of Information and Communication Engineering*, vol. 27, no. 10, pp. 1208–1213, Oct. 2023

RESEARCH EXPERIENCE

Research Mentee

Remote

IonQ, Advised by Dr. Denny Dahl

08/2025–09/2025

- Addressed the Instant Insanity problem on NISQ devices by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods.
- Applied a Variational Quantum Algorithm to the Four Corners Map Coloring problem.

Undergraduate Researcher

University Park, PA

Penn State CSE, Advised by Prof. Kiwan Maeng

08/2024–09/2025

- Conducted in-depth GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differentially private training.
- Proposed an optimization that resolved the dominant Geometric operation overhead, reducing it from 50% to 19% while preserving Gaussian noise quality.

Undergraduate Research Intern**Suwon, South Korea***ArchLab @ SKKU, Advised by Prof. Dongmoon Min*

07/2025–08/2025

- Built foundational expertise in Quantum / Cryogenic Computer Systems through literature reviews, technical presentations and academic defenses.
- Explored quantum computer systems, focusing on superconducting qubits, and cryogenic computer architectures covering processors, interconnects, caches, and memory.

Undergraduate Research Intern**Pohang, South Korea***POSTECH QCQN (Quantum Computing and Quantum Networks)*

01/2025

- Performed 397 nm fluorescence (Doppler) and 729 nm quadrupole spectroscopy on Ca^+ ions and enhanced system stability through targeted adjustments.
- Implemented micromotion optimization via 2D scanning with DC compensation rods.

REFERENCES

Kiwan Maeng, Ph.D. (Advisor)*Assistant Professor, Computer Science and Engineering, The Pennsylvania State University**Email: kvm6242@psu.edu***Denny Dahl, Ph.D.** (Advisor)*Customer Success Director, IonQ**Email: denny.dahl@ionq.co***HONORS & AWARDS**

Quantum Hackathon – Director’s Award

2025

President’s List (Honor Society at SKKU, **2 consecutive years**)

2024, 2025

Dean’s List (**5 consecutive semesters**)

SP2023, FA2023, SP2024, FA2024, SP2025

Korea-U.S. Student Exchange Scholarship in the field of high-tech industry (\$9,000)

FA2024

TEACHING EXPERIENCE

Academic Tutor**Suwon, South Korea***Sungkyunkwan University (SKKU), Intro to Electromagnetism**Fall 2023***International Summer School Teaching Assistant****Seoul, South Korea***Sungkyunkwan University (SKKU), Engineering Mathematics 2**Summer 2023***TECHNICAL SKILLS**

Software: Qiskit, CUDA, C, Python, MATLAB, LaTeX**Hardware:** System Verilog, Verilog, Verdi, Virtuoso, Nsight Systems, Nsight Compute**CERTIFICATION**

Qiskit Global Summer School 2025 – Quantum Excellence, IBM Quantum